Darmanis 2017 GBM — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

4 patients intermix on one continuous manifold (iLISI 2.68 / 4 vs 1.16 pre-Harmony; θ=10, 20 PCs); radius-graph finds 0 detached islands; all 3 sub-lobes reproduce the embryonic relationships.

Gate 2 — module geography → GO (non-branching)

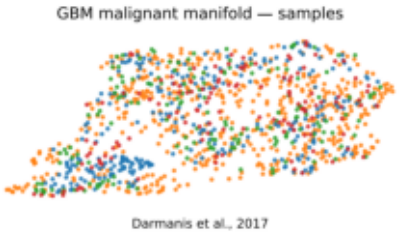
Gast & pSom co-vary spatially (r = +0.72); Crest is the inverse terminal gradient (r = −0.92 vs Gast, −0.59 vs pSom; Crest-high decile at pseudotime 0.98). Single (Gast/pSom)→Crest axis — non-branching case, passes, no bifurcation.

Gate 3 — cross-repression → NA

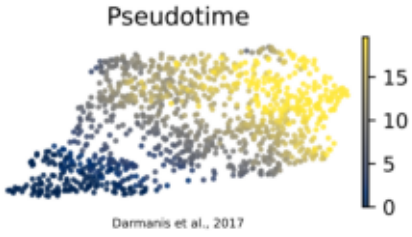
Non-branching manifold: a principal tree shows no Gast-vs-pSom arm divergence (all tips pSom-high, Gast-low), so there is no hub to test. Gate 3 is branching-only — NA.

STEP FIGURES

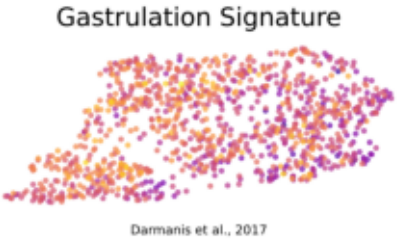
Manifold — samples



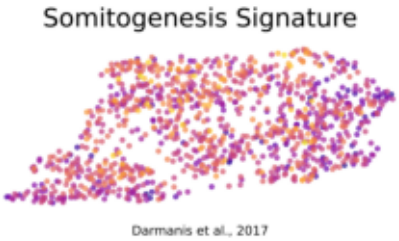
Pseudotime



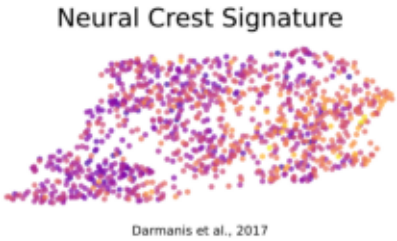
Gastrulation



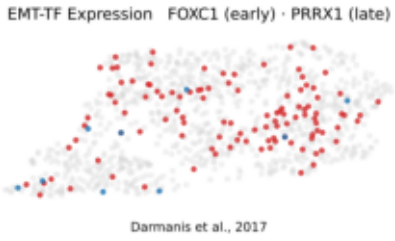
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Glioblastoma (Darmanis et al., 2017), SmartSeq2 full-length scRNA-seq — 1,091 malignant cells across 4 patients. Non-branching Linear-pSom/Crest.

Verdict.
Pattern reproduces as a single non-branching embryonic axis: a Gast/pSom-high origin resolving into a neural-crest-high terminus. The curve is fit on the integrated PCA representation so the axis reaches the Crest pole, not the pSom edge of the 2-D UMAP. Geometry Linear-pSom/Crest (pSom the globally strongest module). EMT-TFs FOXC1 (early/root) → PRRX1 (late/terminus).

Parameter choices

Sample threshold	all 4 kept (≥100-cell floor; no clean low-end gap)	**HVG selection**	scanpy Seurat top-2000 (pagoda2 overdispersion degenerates on pre-log TPM)
Integration	Harmony on sample θ=10, 20 PCs (iLISI plateaus after θ=5; 1.16→2.68/4)	**Trajectory**	ElPiGraph curve on X_pca_harmony (2 tips, 0 forks, 2.97); non-branching
Root	curve tip max Gast+pSom−Crest (developmental origin; neural crest late)	**Geometry**	Linear-pSom/Crest (Gast μ +0.010 · pSom μ +0.099 · Crest μ +0.041)
EMT-TFs	FOXC1 (early/root) → PRRX1 (late/terminus); 10-factor panel		